Tianzhi Li

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Basic Information

I am an incoming **Shun Hing Postdoctoral Fellow** at the Chinese University of Hong Kong (CUHK), working with Prof. Zhongyu Li. I received my Ph.D. in Mechanics & Control from Peking University (PKU) in June 2026, working with Prof. Jinzhi Wang and Prof. Zhisheng Duan. From Dec-2024 to Dec-2025, I was a visiting PhD at Nanyang Technological University (NTU), Singapore, advised by Prof. François Gay-Balmaz. Prior to that, I received my B.Sc. in Mathematics from Beijing Institute of Technology (BIT) in 2021 under the supervision of Prof. Donghua Shi.

My research interest lies at the intersection of **Geometric Mechanics & Control**, **Physics-Informed Learning**, and **Stochastic Dynamics** for robotic and mechanical systems. I am broadly interested in leveraging physical principles and differential geometry to develop computationally efficient algorithms for **control**, **estimation**, and **learning** with sound theoretical guarantees.

Academic Experience

- **Peking University** — Sep 2021 - Jun 2026
  PhD (Dynamical Systems and Control)
- **Nanyang Technological University**, Singapore — Dec 2024 - Dec 2025
  Visiting PhD (Geometric Mechanics)
- **Beijing Institute of Technology** — Sep 2017 - Jun 2021
  Bachelor (Mathematics)

Teaching Experience

- Teaching Assistant, "Ordinary Differential Equations", Peking University — Fall 2023
  Undergraduate-level course, covering more than 150 students
  Recieved **Outstanding Teaching Assistant Award**
- Teaching Assistant, "Linear Algebra", Peking University — Spring 2023
  Undergraduate-level course, covering more than 150 students
- Teaching Assistant, "Linear Algebra", Peking University — Fall 2022
  Undergraduate-level course, covering more than 150 students
  Delivered Lecture 5 "Determinants"
  Recieved **Outstanding Teaching Assistant Award**
- Teaching Assistant, "Analytic Mechanics", Peking University — Spring 2022
  Undergraduate-level course, covering more than 100 students
- Teaching Assistant, "Ordinary Differential Equations", Peking University — Fall 2021
  Undergraduate-level course, covering more than 150 students

Publications

Preprints

[P1] Y. Lin, Z. Duan, **T. Li**, B. Wu, and Z. Sun, On the Robustness in Data-Driven Nonlinear Optimal Control: From Stability to Optimality, under review.

[P2] M. Du, F. Gay-Balmaz, **T. Li**, and D. Shi, Stochastic Geometrically Exact Beam: Continuous-Time Geometric Principles and Variational Discretizations, to be submitted.

Published Papers

[1] **T. Li** and J. Wang, Variational Unscented Kalman Filter on Matrix Lie Groups, **Automatica** (Regular Paper), 172: 111995, 2025. [Paper Link]

[2] **T. Li**, R. Fu, and J. Wang, Reduced Dynamics and Geometric Optimal Control of Nonequilibrium Thermodynamics: Gaussian Case, **Automatica** (Regular Paper), 164: 111626, 2024. [Paper Link]

[3] **T. Li** and J. Wang, Physics-Informed Gaussian Process Learning on Lie Groups, **Journal of Guidance, Control, and Dynamics**, 48 (11), pp. 2654-2662, 2025. [Paper Link]

[4] **T. Li**, J. Wang, and Z. Duan, Structure-Preserving Unscented Kalman Filter for Planar Mobile Robots, **IEEE Control Systems Letters**, vol. 9, pp. 2157-2162, 2025. [Paper Link]

[5] **T. Li**, F. Gay-Balmaz, D. Shi, and J. Wang, Variational Principle for Stochastic Nonholonomic Systems **Part II**: Stochastic Nonholonomic Integrator. In: Nielsen, F., Barbaresco, F. (eds) Geometric Science of Information (**GSI'25**), Saint-Malo, France, vol. 16034, pp. 225-233. Springer, 2026. [Paper Link]

[6] **T. Li**, F. Gay-Balmaz, D. Shi, and J. Wang, Variational Principle for Stochastic Nonholonomic Systems **Part I**: Continuous-Time Formulation. In: Nielsen, F., Barbaresco, F. (eds) Geometric Science of Information (**GSI'25**), France, vol. 16034, pp. 204-213. Springer, 2026. [Paper Link]

[7] **T. Li** and J. Wang, A Structure-Preserving Learning Scheme on SO(3), 2024 43rd IEEE Chinese Control Conference (**CCC'24**), Kunming, China, 2024, pp. 5149-5152. [Paper Link]

[8] **T. Li** and J. Wang, Multisymplectic Unscented Kalman Filter for Geometrically Exact Beams. In: Nielsen, F., Barbaresco, F. (eds) International Conference on Geometric Science of Information (**GSI'23**). Lecture Notes in Computer Science, Saint-Malo, France, vol. 14072, pp. 60-68, Springer Verlag. [Paper Link]

[9] **T. Li** and J. Wang, A Physics-Informed Gaussian Process Regression Algorithm for The Dynamics of The Planar Pendulum, 2023 42nd IEEE Chinese Control Conference (**CCC'23**), Tianjin, China, pp. 5163-5167, 2023. [Paper Link]

[10] **T. Li** and J. Wang, A Statistical Dynamical Algorithm for Gaussian Multi-Agent Systems Under Hamel's Formalism, 34th IEEE Chinese Control and Decision Conference (**CCDC'22**), Hefei, China, pp. 1344-1349, 2022. [Paper Link]

Honors and Awards

- **National Scholarship** (top 1%) — Sep-2025
  Issued by Ministry of Education, China
- **Merit Student of Beijing** — Dec-2025
  Issued by Beijing Education Commission
- **Presidential Doctoral Scholarship** — Jun-2025
  (The most prestigious scholarship issued by Peking University)
- **China Scholarship Council Visiting PhD Scholarship** — Jul-2024
  Issued by China Scholarship Council
- **Merit Student Award** (top 1%) — Sep-2025
  Issued by Peking University
- **Excellent Academic Research Award** (top 1%) — Sep-2025
  (only 2 awardees at SAMR, Peking University)
- **Dean's Doctoral Scholarship - First Prize** — Oct-2024
  Issued by College of Engineering, Peking University
- **Yuehua Luo Scholarship** — Nov-2024
  Issued by Peking University
- **Outstanding Teaching Assistant Award** ($\times$ 2) — Mar-2024 & Apr-2023
  Issued by College of Engineering, Peking University

Conference Talks & Invited Talks

- Variational Principle for Stochastic Nonholonomic Systems Part II: Stochastic Nonholonomic Integrator
  The International Conference on Geometric Science of Information (GSI)
  Saint-Malo, France, Oct-2025
- Variational Principle for Stochastic Nonholonomic Systems Part I: Continuous-Time Formulation
  The International Conference on Geometric Science of Information (GSI)
  Saint-Malo, France, Oct-2025
- A Structure-Preserving Learning Scheme on SO(3)
  The 43rd Chinese Control Conference (CCC)
  Kunming, China, Jul-2024
- Multisymplectic Unscented Kalman Filter for Geometrically Exact Beams
  International Conference on Geometric Science of Information (GSI)
  Saint-Malo, France, Aug-2023
- A Physics-Informed Gaussian Process Regression Algorithm for the Dynamics of the Planar Pendulum
  The 42nd Chinese Control Conference (CCC)
  Tianjin, China, Jul-2023
- A Statistical Dynamical Algorithm for the Dynamics of Gaussian Distributions
  The 34th Chinese Control and Decision Conference (CCDC)
  Hefei, China, Aug-2022

Academic Service

Journal Reviewer: Automatica, International Journal of Dynamics and Control.

Conference Reviewer: IEEE Conference on Decision and Control (CDC), IFAC Workshop on Lagrangian and Hamiltonian Methods for Nonlinear Control (LHMNC), International Symposium on Mathematical Theory of Networks and Systems (MTNS).

References

Professor Zhisheng Duan
Position: Professor of Dynamics and Control
Deputy Director of State Key Laboratory for Turbulence & Complex Systems
Affiliation: Peking University
E-mail: duanzs@pku.edu.cn

Professor François Gay-Balmaz
Position: Associate Professor of Mathematics
Affiliation: Nanyang Technological University
E-mail: francois.gb@ntu.edu.sg

Professor Jinzhi Wang
Position: Professor of Dynamics and Control
Affiliation: Peking University
E-mail: jinzhiw@pku.edu.cn

Professor Donghua Shi
Position: Professor of Mathematics
Director of Beijing Key Laboratory on MCAACI
Affiliation: Beijing Institute of Technology
E-mail: dshi@bit.edu.cn